Reproducibility Report: Coppock, Leeper and Mullinix (2018)

Generalizability of Heterogeneous Treatment Effect Estimates Across Samples

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This repository holds the actively maintained replication code for Coppock, Leeper and Mullinix (2018), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1073/pnas.1808083115
Replication archive	10.7910/DVN/4WNGEJ
Source of the 16 Mullinix et al. study pairs	10.1017/XPS.2015.19
Source of the 11 Coppock study pairs	10.1017/psrm.2018.10
Pre-analysis plans	None; the paper is a reanalysis of existing experiments

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the directory each file belongs in, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all 46 files. They do not always, so the script verifies against the served bytes and reports any disagreement. It also refuses to proceed if `original/` holds anything the deposit does not, which is how a directory quietly acquires output from a run rather than from the archive. Either way the exact bytes are pinned in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything, plus `in_text_claims.R`, which recomputes every quantity the article states in prose and prints it beside the sentence that states it. `ground_truth/` ties every published number to the code that produces it: the comparison table itself, the appendix values as read off the published PDF, and `published_claims.csv`, the token-by-token extraction of the article and appendix that decides what coverage means. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`; `coppock_leeper_mullinix_2018_errata.pdf` at the root corrects the published article.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_leeper_mullinix_2018.Rproj`, and run:

```r
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its 46 checksums, and produces every table and figure into `maintained/output/`. It takes about half a minute. Individual scripts can be run on their own, in the order `run_all.R` uses, since each reads its inputs from `maintained/output/`.

Required packages: tidyverse, estimatr, broom, deming, lmtest, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and every file in it comes back byte-identical, the two figure PDFs included. A PDF ordinarily records the time it was written, which would make those two differ on every run for no reason a reader cares about, so `run_all.R` blanks the embedded `/CreationDate` and `/ModDate` before the ground truth is built.

1 Paper Overview

Citation: Coppock, A., Leeper, T. J. and Mullinix, K. J. (2018). Generalizability of heterogeneous treatment effect estimates across samples. *Proceedings of the National Academy of Sciences*, 115(49), 12441–12446.

Research question: Sample average treatment effects estimated on nationally representative samples and on online convenience samples correspond closely. Two things could explain that: treatment effects are largely homogeneous, so who is in the sample does not matter, or they are heterogeneous in ways unrelated to who selects into a convenience sample. Which is it?

Design: A reanalysis of 27 pairs of survey experiments, each run once on a nationally representative GfK/TESS sample and once on Amazon Mechanical Turk under an identical protocol. Sixteen pairs come from Mullinix, Leeper, Druckman and Freese (2015) and eleven from Coppock (2018). Within each version of each study, conditional average treatment effects are estimated by difference-in-means for 16 subgroups defined by age, education, gender, ideology, partisanship and race, coarsened to at most three categories each, with HC2 standard errors. The original and replication CATEs are then compared in three ways: a Deming regression of one on the other, which is an errors-in-variables fit that takes the estimated standard errors on both axes and gets its confidence interval from the jackknife; a joint F test against the null of no difference in the pattern of heterogeneity by sample; and a direct significance comparison of each pair of CATEs.

Main finding: Across 393 subgroup comparisons the correspondence is strong. The across-study Deming slopes run from 0.71 to 1.01, all positive, and all but one confidence interval covers 1. The difference between the two versions is significant 59 times, 15 per cent, and in no case do the two point in opposite directions while both are distinguishable from zero. Within studies, the CATEs cluster tightly around the study's overall effect, so there is little heterogeneity for sample composition to interact with. The conclusion is that treatment effect homogeneity, rather than heterogeneity orthogonal to selection, explains why convenience samples give answers so close to representative ones.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited, and the reason is entirely deprecation. Five of its eight scripts stop with an error in a clean R session, and every one of the five is a one-line fix that no published number depends on.

Table 1: Every way the deposited archive fails in a current R session

Script	Failure	Fix
CLM_study_labels.R	‘write_rds(x, path =)‘	‘file = ‘
CLM_estimate_CATEs.R	‘write_rds(x, path =)‘	‘file = ‘
CLM_f_tests.R	‘write_rds(x, path =)‘	‘file = ‘
CLM_appendix.R	‘read_rds(path =)‘	drop the argument name
CLM_tables.R	column dropped by an earlier ‘transmute()‘	move the diagnostic line above it

The `path` argument of `read_rds()` and `write_rds()` was soft-deprecated in readr 1.4 and is now gone: `write_rds()` reports a missing `file` argument and `read_rds()` reports an unused one. The fifth failure is different in kind and worth naming, because it is the shape of bug that only ever appears once someone else runs the code. `CLM_tables.R` overwrites `group_deming` with a `transmute()` that keeps three formatted columns, then asks the overwritten object for the minimum and maximum of `estimate`, a column that no longer exists. In the session it was written in, `group_deming` still held the estimate; from a clean start it does not. The line is a diagnostic that prints an in-text number and nothing downstream depends on it, so moving it two lines earlier is the whole repair.

With those five edits the archive reproduces itself exactly. Running all eight scripts in order from a directory holding only the deposited data files and scripts regenerates `group_table.tex`, `study_table.tex` and `appendix_tables.tex` byte for byte against the deposited copies, and regenerates the two derived data objects, `CLM_scatter_df.rds` and `CLM_f_tests_df.rds`, to within 1e-13. Nothing here is stochastic. The archive calls no random number generator at all, so the sampler change in R 3.6 does not touch it, and the two `set.seed(343)` calls in `CLM_tables.R` have no effect: `deming()` gets its confidence interval from the jackknife, which is deterministic, and returns the same slope, standard error and interval at any seed.

Two smaller observations. The deposited `CLM_results.rds` was saved as a grouped data frame in a format dplyr dropped in version 0.8, so current dplyr refuses to group it until it has been passed through `ungroup()`; the object's contents are intact and the archive's own scripts never hit the problem because they regenerate it. And `CLM_figures.R` calls `library(coefplot)` without ever using the package, which matters only because that call is what attaches ggplot2, the package it does use.

2.2 Does the maintained rewrite reproduce the paper?

Every number the article prints reproduces from the deposited code. Two columns of the appendix tables are wrong at source, and the rewrite reports the correct quantity rather than carrying the error forward.

Table 2: Reproduction verdict by component

Component	Verdict
Table 1 (27 study pairs, 8 cells each)	216 of 216 cells reproduce
Table 2 (16 subgroups, 5 cells each)	80 of 80 cells reproduce
Figures 1 and 2	All 393 plotted comparisons reproduce
Appendix Tables 1 to 27, estimates	All 3,935 CATE, SE, p-value and interval cells reproduce
Appendix Tables 1 to 27, N column	Reproduces; 98 of 787 cells are one to three observations short
Appendix Tables 1 to 27, Prop column	Reproduces; every one of the 787 is the share divided by the number of regression terms
In-text quantities	46 prose claims, every one covered by a block

The ground truth records 425 claims, 410 of which carry a value the article or its appendix prints. **All 410 are reproduced by the deposited code.** 379 are also reproduced by the rewrite; the remaining 31 are the N and Prop columns of the appendix tables, where the rewrite deliberately reports the correct quantity instead. Those two corrections are set out under Errata below.

Nothing in the paper's argument turns on either column. The estimates, standard errors, p-values and confidence intervals in the same tables are untouched, both main tables are untouched, and both figures are untouched.

One further finding concerns not a number but what a number counts, and it is entry 2 of the errata. Table 1's `Original N` and `Mturk N` columns, which sum to the abstract's 101,745 individual survey responses, mean different things in the two halves of the table. For the eleven studies from Coppock (2018) they are the observations the estimates use. For the sixteen from Mullinix et al. (2015) they are the number of rows in the deposited file, which is the whole survey wave rather than the respondents assigned to that experiment, and the same MTurk waves are counted once per study that drew on them. Bergan (2012) is listed at 1,206 and 1,913; the subgroup estimates use 396 and 587. The figure is a faithful description of the deposit and reproduces exactly, and the paper's claim about it is that Table 1 gives "the sample sizes used in the analyses reported here". For the sixteen it does not. `maintained/output/study_ns.csv` carries both counts for every study.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, doi 10.7910/DVN/4WNGEJ, 46 files in a single `replication_archive/` directory: 27 data files, 8 R scripts, and 11 output files the scripts themselves produce.

The deposit ships a README naming every data file and pairing each script with the output it writes. Both halves of that description are accurate, which is not the norm: all 27 data files are present under the names given, all 8 scripts exist, and each writes what the README says it writes.

The dependency chain runs `CLM_study_labels.R` and `CLM_study_summaries.R` first, then `CLM_estimate_CATEs.R`, which does the estimation and saves the two objects everything downstream reads, then `CLM_f_tests.R`, and finally the three scripts that produce output: `CLM_figures.R`, `CLM_tables.R` and `CLM_appendix.R`. `CLM_in_text_figures.R` prints the paper's in-text quantities to the console without writing anything, so those numbers exist in the deposit only as the console output of a script nobody has run.

Because the deposit also ships the intermediate objects, three of its scripts appear to succeed on a first run even though the scripts that produce their inputs have failed. Running `CLM_figures.R` in the deposit as downloaded produces both figures without error, using the deposited `CLM_scatter_df.rds` rather than one this run computed. The distinction is invisible unless the directory is stripped back to data and code, which is how the reproduction above was done.

Two composition risks that bite other archives of this vintage do not bite this one. `lmtest::waldtest()` is called on `lm_robust` objects 54 times, and current `estimatr` exports a `vcov` method that `waldtest()` will find where it once fell through to a classical F test; here the regenerated p-values match the deposited ones to 6e-14, so the test being performed has not changed. And the packages the archive leans on are all still maintained: `deming` was last released in 2024, `reshape2` and `coefplot` in 2025.

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4 Errata

The maintained rewrite corrects two columns of the appendix tables. Neither correction changes an estimate, and neither touches the main text.

The N column is the residual degrees of freedom plus two. `CLM_appendix.R` sets `n = df + 2`, which is the number of observations only when the regression has exactly two terms, an intercept and treatment. Twenty-three of the 27 studies do, and their N columns are right. Four do not, because their subgroup regressions carry an extra regressor: Transue (2007) adds a blocking factor, Nicholson (2012) adds party identification for the subgroups not defined by party, McGinty, Webster and Barry (2013) adds a policy indicator and Hiscox (2006) adds a valence indicator. Those four lose one, one, two and three observations per cell respectively, 98 cells in all. The rewrite reports the number of observations the fit actually used.

The Prop column is the share divided by the number of regression terms. The appendix says the column "describes what proportion of the subjects in a given experiment belong to the associated covariate class", and the three age shares in a table should therefore sum to one. In the published tables they sum to one half, or to one third, one quarter or one fifth in the four studies above. The denominator is a sum taken over every row of the fitted results rather than over the treatment coefficients alone, so each subgroup is counted once per regression term. The rewrite divides by the sum over subgroups.

Both corrections are recorded in the ground truth as `match_rewrite = 0` with `defect_locus = archive`, and `maintained/output/table_a_cate_estimates.csv` carries the deposit's own version of each column alongside the corrected one, so the published values remain checkable. Under the deposit's formulas all 787 N cells and all 787 Prop cells reproduce exactly.

A note correcting the article. Two sentences describe a column of a table incorrectly, and a sentence a reader would quote and be misled by is a correction to the published record rather than a note about the deposit. Both are set out in `coppock_leeper_mullinix_2018_errata.pdf` at the root of this repository, with their numbers computed from the pipeline when the document is rendered. Neither changes a conclusion, and they are numbered here as the note numbers them. **Entry 1** is the Prop defect above: the appendix tells the reader what that column contains and the sentence is not true of the numbers printed beneath it.

Entry 2 is the Table 1 sentence set out above, that Table 1 gives "the sample sizes used in the analyses reported here" when its columns count the whole survey wave. It was recorded here as an observation about what a number counts before it was recognised as a correction to the published record, which is the same shape as entry 1: a sentence telling the reader what a column contains, and not true of the column beneath it.

Entries 3 and 4 are in the reference list, and no ground truth row reaches them: every printed entry was sent whole to Crossref and the authoritative record checked back into it. References 1 and 2 both print “Druckman JN, Green DP, Kuklinski JH, Arthur L”, where the fourth author is Arthur Lupia and his given name has been promoted to a surname. No citation in the text points at the wrong work.

Every published estimate still reproduces, and the one internal contradiction is that sentence. Every number the article and its appendix print reproduces from the deposited code at the precision the page carries. Exactly one ground truth row is `paper_internal`, the Table 1 sentence: the article contradicts itself there rather than merely being imprecise, since Table 1 reports 1,206 for Bergan (2012)’s original sample where every attribute in the corresponding panel of appendix Table 1 sums to 396. Nowhere does it contradict its own data. Of the descriptive claims that assert something about the estimates without stating a number, all hold but the two corrected here. The two shapes of erratum that no failing row would ever surface were looked for and are absent: no sentence states the cause of an error elsewhere in the paper, and no published number agrees with the deposited code while the article misdescribes the method that produced it. The one description worth flagging and not correcting is “we estimate all CATEs via difference-in-means”, which is exact for 23 of the 27 study pairs and, for the other four, describes a fit that also carries another arm of the study’s own factorial design or the party identification its assignment probabilities depend on.

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5 Ground Truth

`ground_truth/coppock_leeper_mullinix_2018_ground_truth.csv` has 425 rows. Every published float is covered, and the appendix is covered cell by cell.

Table 3: Ground truth coverage

Component	Claims	With a published value	Archive reproduces	Rewrite reproduces
Table 1	216	216	216	216
Table 2	80	80	80	80
Appendix Tables 1 to 27	81	81	81	50
In-text quantities	46	33	33	33
Figures 1 and 2	2	0	0	0

`value_paper` comes only from the published documents. Table 1 and Table 2 were transcribed from the typeset pages and cross-checked against a rendered image of each page rather than a text extraction, since a multi-column table read as text is easy to mis-map by one column. The 5,509 appendix cells were read off the appendix PDF into `ground_truth/published_appendix_values.csv`, which is committed, and a rendered page was checked against the parse. `value_script` is read out of the deposit's own `study_table.tex`, `group_table.tex` and `appendix_tables.tex` and out of its two derived data objects. `value_rewrite` is read out of `maintained/output/`. No published number is an input to any computation in `maintained/`.

Of the 425 rows, 15 carry no published value. Two are Figures 1 and 2, which print no numbers. The other 13 are sentences that describe the estimates without stating a quantity, and they divide in two. Eight of them reduce to a truth value and carry one: that the across-study slopes are all positive, that the interval excluding 1 is the conservative group's, that the within-study slopes take both signs, that the standard error of the difference is the root sum of squares the footnote gives, that the tests use a normal approximation, that the appendix p-values are two-sided, and that the appendix's Prop column is the share it says it is. All but the last hold. The remaining 5 describe a shape rather than a threshold, so the block prints the distribution and reaches no verdict: "mostly uncorrelated", "smaller than the across-study slopes", "tightly clustered", "only limited evidence", and "all CATEs via difference-in-means". Each is checked against the study-level estimates rather than against a pooled figure. The first of them is a claim about 27 correlations rather than one:

Table 4: Correlation of original and replication CATEs

Quantity	Value
Lowest within-study correlation	-0.63
Median within-study correlation	0.24
Highest within-study correlation	0.87
Study pairs with a positive correlation	17 of 27
Study pairs above 0.50	5 of 27
All 393 comparisons pooled	0.72

The characterisation holds. The within-study correlations are centred near zero and scatter widely on either side, which is what "mostly uncorrelated" should mean, and the contrast with the pooled figure of 0.72 is exactly the paper's point: across studies the CATEs correspond closely, within a study they barely move.

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6 Maintained Rewrite

Ten scripts in `maintained/`, one shared `helpers.R`, and every output in `maintained/output/`.

Table 5: The maintained rewrite

Script	Produces
helpers.R	Packages, study labels, subgroup labels, and the CATE and Deming helpers
estimate_cates.R	cate_estimates.csv, cate_scatter.csv
f_tests.R	f_tests.csv
study_ns.R	study_ns.csv
table_1_within_study_correspondence.R	table_1_within_study_correspondence.csv
table_2_across_study_correspondence.R	table_2_across_study_correspondence.csv
table_a_cate_estimates.R	table_a_cate_estimates.csv
figure_1_across_study_correspondence.R	figure_1_across_study_correspondence.pdf, .png, .csv
figure_2_within_study_correspondence.R	figure_2_within_study_correspondence.pdf, .png, .csv
text_correspondence_summary.R	text_correspondence_summary.csv
text_within_study_correlation.R	text_within_study_correlation.csv

The substitutions the rewrite makes:

Table 6: Substitutions

Original	Replacement	Reason
‘gather()‘	‘pivot_longer(values_transform =)‘	Retired. The covariate columns are a mix of factor and numeric across studies, which ‘gather()‘ coerced silently with a warning; the transform makes the coercion explicit
‘do(tidy(...))‘	‘reframe(tidy(...))‘	‘do()‘ is superseded
‘reshape2::melt()‘ and ‘dcast()‘	‘pivot_wider()‘	reshape2 is superseded
‘data_frame()‘	‘tibble()‘	Deprecated since tibble 1.1
‘write_rds(path =)‘, ‘read_rds(path =)‘	Write CSV to ‘output/‘	The argument is gone, and a text output can be diffed
‘xtable()‘ plus ‘sink()‘	‘write_csv()‘	‘sink()‘ writes to a hardcoded path and cannot be checked; a CSV can be compared cell by cell
‘library(coefplot)‘	‘library(ggplot2)‘	The archive used coefplot only as a way of attaching ggplot2

‘geom_segment()‘ for interval bars	‘geom_linerange()‘	The current idiom for a confidence interval on a scatter

`deming::deming()` is kept. It is the only implementation of the estimator the paper uses, it is actively maintained, and its jackknife interval is what the paper describes.

The rewrite also adds three things the deposit does not carry. `f_tests.csv` records the F statistic and both degrees of freedom rather than the p-value alone, so the test is inspectable. `study_ns.csv` records the observations that enter the estimates alongside the sample sizes Table 1 prints. And each figure script writes a CSV of the values it plots, because two PDFs never compare equal and a figure that writes no numbers cannot be checked at all.

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7 Table 1: Within-Study Correspondence

Table 7: Table 1 as the rewrite produces it. Every cell matches the published table.

Study	Original N	MTurk N	Slope (SE)	95% CI	N comparis
Bergan (2012)	1206	1913	0.75 (0.20)	[0.37, 1.12]	
Brader (2005)	280	1709	3.56 (1.68)	[-30.74, 37.86]	
Brandt (2013)	1225	3131	4.49 (1.96)	[-6.25, 15.23]	
Caprariello and Reis (2013)	825	2729	-4.38 (2.00)	[-7.83, -0.93]	
Chong and Druckman (2010)	958	1400	0.17 (0.18)	[-0.58, 0.92]	
Craig and Richeson (2014)	608	847	-0.95 (0.36)	[-1.56, -0.34]	
Denny (2012)	1733	1913	2.83 (1.04)	[1.19, 4.47]	
Epley et al. (2009)	1019	1913	0.68 (0.64)	[-2.52, 3.88]	
Flavin (2011)	2015	2729	0.23 (0.20)	[-0.15, 0.62]	
Gash and Murakami (2009)	1022	3131	2.78 (1.01)	[1.59, 3.96]	
Hiscox (2006)	1610	2972	2.50 (1.07)	[0.94, 4.07]	
Hopkins and Mummolo (2017)	3266	2972	-1.84 (0.85)	[-4.06, 0.37]	
Jacobsen, Snyder and Saultz (2014)	1111	3171	-4.73 (1.99)	[-8.32, -1.14]	
Johnston and Ballard (2016)	2045	2985	0.13 (0.53)	[-0.26, 0.53]	
Levendusky and Malhotra (2015)	1053	1987	-0.16 (0.35)	[-1.50, 1.19]	
McGinty, Webster and Barry (2013)	2935	2985	2.53 (1.10)	[1.09, 3.97]	
Murtagh et al. (2012)	2112	3131	0.34 (0.34)	[-0.20, 0.88]	
Nicholson (2012)	781	1099	-23.05 (16.21)	[-396.69, 350.58]	
Parmer (2011)	521	3277	1.71 (0.75)	[-0.12, 3.54]	

Pedulla (2014)	1407	1913	-57.93 (17.90)	[-363.42, 247.55]
Peffley and Hurwitz (2007)	905	1285	2.23 (1.17)	[-1.08, 5.54]
Piazza (2015)	1135	3171	-2.15 (0.74)	[-3.83, -0.47]
Shafer (2017)	2592	2729	-24.13 (9.75)	[-162.31, 114.05]
Thompson and Schlehofer (2014)	591	3277	0.24 (0.60)	[-1.08, 1.56]
Transue (2007)	345	367	-1.67 (1.02)	[-7.52, 4.17]
Turaga (2010)	774	3277	1.44 (0.54)	[-0.86, 3.74]
Wallace (2011)	2929	2729	4.74 (1.86)	[-7.96, 17.43]

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8 Table 2: Across-Study Correspondence

Table 8: Table 2 as the rewrite produces it. Every cell matches the published table.

Covariate class	Slope (SE)	95% CI	N comparisons
Age: 18 - 39	0.82 (0.08)	[0.51, 1.12]	27
Age: 40 - 59	0.86 (0.08)	[0.55, 1.17]	27
Age: More than 60	0.95 (0.13)	[0.61, 1.29]	26
Less than College	0.93 (0.06)	[0.61, 1.25]	26
College	0.87 (0.10)	[0.46, 1.29]	26
Graduate School	0.72 (0.13)	[0.28, 1.15]	26
Men	0.87 (0.07)	[0.49, 1.25]	26
Women	0.91 (0.07)	[0.61, 1.20]	26
Liberal	0.71 (0.11)	[0.37, 1.05]	20
Moderate	1.01 (0.11)	[0.49, 1.53]	20
Conservative	0.75 (0.09)	[0.52, 0.99]	20
Democrat	0.88 (0.07)	[0.50, 1.26]	24
Independent	0.94 (0.16)	[0.19, 1.68]	23
Republican	0.86 (0.08)	[0.63, 1.08]	24
Nonwhite	0.95 (0.11)	[0.45, 1.46]	25
White	0.92 (0.06)	[0.66, 1.18]	27

The one interval that excludes 1 is Conservative, at [0.52, 0.99], which is what the paper says.

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9 Figure Verification

Neither figure prints a number, so the check is against the values plotted. Both figure scripts write the coordinates of every point and every interval limit to CSV, and all 393 comparisons agree with the deposit's own `CLM_scatter_df.rds` to better than 1e-12.

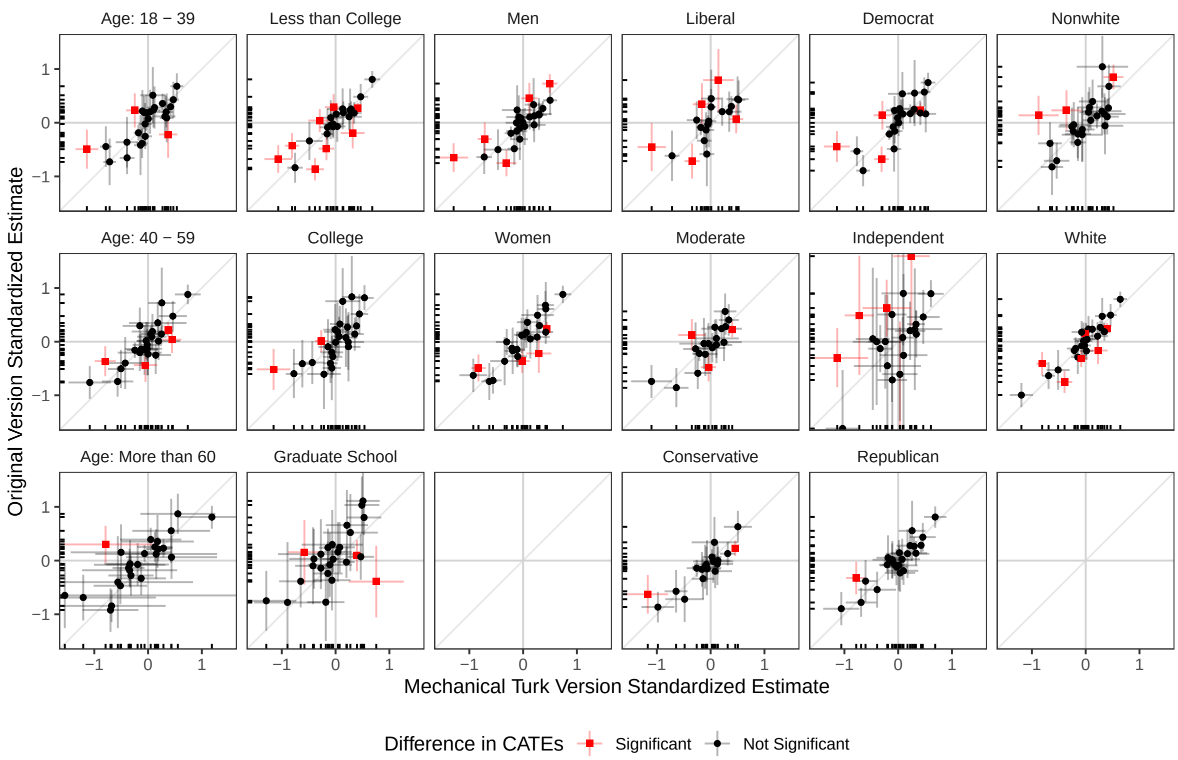

Figure 1: Figure 1: across-study correspondence of CATEs, as the rewrite produces it.

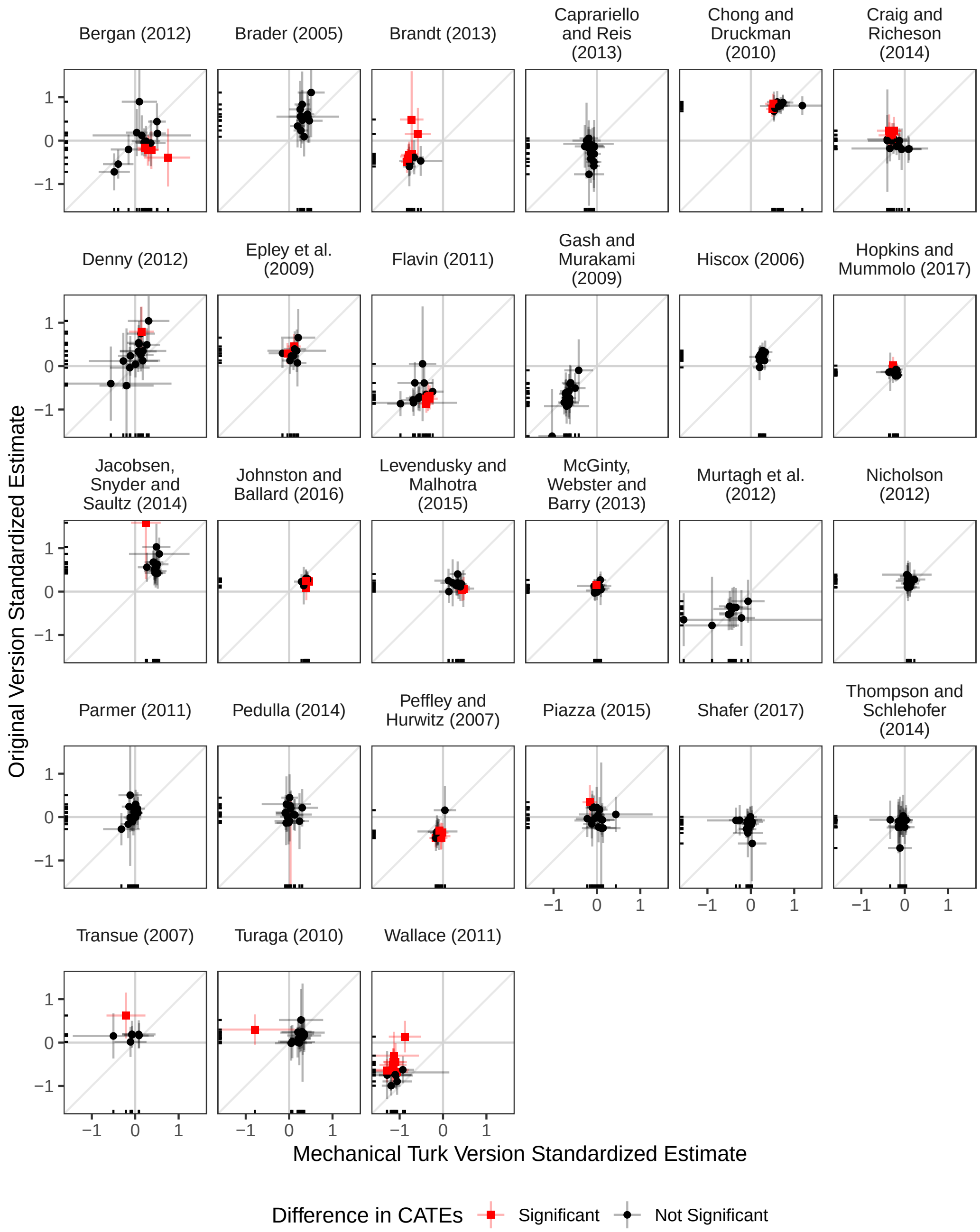

Figure 2: Figure 2: within-study correspondence of CATEs, as the rewrite produces it.

10 In-Text Quantities

Table 9: Every quantity the article states in prose

Section	Quantity	Published	Rewrite	Match
Abstract	Study pairs analysed	27	27	yes
Abstract	Individual survey responses	101745	101745	yes
Significance statement	Survey experiments replicated	27	27	yes
Significance statement	Individual survey responses	101745	101745	yes
Methods and Materials	Original-replication pairs reanalysed	27	27	yes
Methods and Materials	Distinct subgroups	16	16	yes
Methods and Materials	Separate experiments	54	54	yes
Methods and Materials	Study pairs, same sentence	27	27	yes
Methods and Materials	Pretreatment attributes	6	6	yes
Methods and Materials	Largest number of categories per attribute	3	3	yes
Methods and Materials	Youngest age band, lower bound	18	18	yes
Methods and Materials	Youngest age band, upper bound	39	39	yes
Methods and Materials	Middle age band, lower bound	40	40	yes
Methods and Materials	Middle age band, upper bound	59	59	yes
Methods and Materials	Oldest age band, lower bound	60	60	yes
Results footnote	Significance threshold for the difference	0.05	0.05	yes
Results	Comparison opportunities	393	393	yes
Results	Difference-in-CATEs significant	59	59	yes
Results	Share of comparisons significant	15	15.01	yes
Results	Sign disagreements with both versions significant	0	0	yes
Results	Comparison opportunities, same sentence	393	393	yes
Results	CATEs significant in the original version	156	156	yes
Results	Of those, significant in the MTurk version	118	118	yes
Results	CATEs indistinguishable from zero in the original version	237	237	yes
Results	Of those, indistinguishable in the MTurk version	158	158	yes
Results	Overall significance match rate	70	70.23	yes
Results	Smallest across-study slope	0.71	0.71	yes
Results	Largest across-study slope	1.01	1.01	yes
Results	Across-study intervals excluding one	1	1	yes
Results	F tests failing to reject at 0.05	25	25	yes
Results	Study pairs offering a joint F test	27	27	yes

Discussion	Pairs of survey experiments	27	27	yes
Appendix table headers	Confidence level of the CATE intervals	95	95.00	yes

The 13 prose claims missing from this table are the descriptive ones listed under Ground Truth above, which state no number. Eight carry a computed truth value and all but one holds; the exception is the appendix's description of its own Prop column, which is the subject of the errata note.

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11 The Extraction and the Two Instruments

The ground truth answers "does the pipeline reproduce what the paper prints?" It cannot answer "did anyone look at every number the paper prints?", because a ground truth built from the pipeline's own outputs is silent about a sentence nobody thought to check. Two further artifacts close that gap.

`ground_truth/published_claims.csv` is the extraction. The article's six pages and the appendix were read token by token rather than searched for patterns: every numeric string in the body, and a second pass for numbers written as words ("six attributes", "three categories", "two nested models"), classified by hand into the five kinds a claim can be. Author footnote markers, postal codes, citation markers, editorial dates, the DOI, page numbers, cross-references of the form "Fig. 1" and the reference lists of both documents are excluded as provably not claims; everything else has a row. The extraction covers the published article and the published supplementary appendix, at sentence granularity for prose and at float granularity for the tables and figures, whose cells the ground truth already compares one by one.

Table 10: The extraction, by claim type

Claim type	Needs a block	Claims
definitional	no	14
definitional	yes	7
descriptive	yes	13
pipeline	yes	26
structural	no	3
transcribed	no	1

Alongside these 64 prose claims the extraction carries 31 rows recording how many numbers each published float prints, 5,805 in all, which the build checks against the transcriptions float by float.

A claim needs a block when a block can print something true. Every `pipeline` and `descriptive` claim does, and so does every definitional claim the pipeline can reach: the age band boundaries come out of the subgroup labels the data carry, the confidence level out of the interval half widths, and the significance threshold out of the point at which the classification of the 393 differences changes. The claims that need no block are the ones nothing in the deposit could confirm or refute, and they are verified where they are used instead: the level of the jackknife interval is `deming()`'s own `conf` argument, `HC2` is `lm_robust`'s default and is visible in `run_cates()`, and the hypothetical match rates of 100 per cent are not estimates at all.

`maintained/in_text_claims.R` is the second instrument. It recomputes each of those claims from `maintained/output/` by a path of its own and prints `CLAIM <id> = <value> || <label>` for each. Where `build_ground_truth.R` reaches a quantity through `text_correspondence_summary.csv`, the claims file goes back to the estimates that summary was built from, so the two derivations are separate and a disagreement means one of them is wrong. Neither file refits anything; estimation happens once, in the analysis scripts, and only derivation is duplicated.

The coverage gate is the last step of `build_ground_truth.R`. It runs the claims file as a program in its own environment, captures its output, and stops the build unless the 46 printed claim identifiers are exactly the 46 the extraction requires and every printed value matches the ground truth's own answer, rendered at the precision the article prints. The precision itself lives only in the extraction, so the two instruments cannot name different ones, and the extraction's stored strings are checked against that precision before anything consumes them. Reading the file as text would not do: a block that errors, or one that computes silently and prints nothing, satisfies a textual check completely.

The gate also reconciles the extraction against the ground truth's own transcription of the same pages, which are two hand readings of one document and nothing else compares them.

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12 Rewrite Verification

The rewrite is deterministic. Running `run_all.R` twice in succession leaves every file in `maintained/output/` byte-identical, the two figure PDFs included, once their embedded write timestamps have been blanked. Nothing in the pipeline draws a random number, so there is no seed to set and no sampler version to pin.

The estimates themselves were checked against the deposit rather than only against the paper, which is a stronger test because the deposit carries more digits than the printed tables do:

Table 11: Verification of the rewrite

Check	Result
1,768 CATE rows against CLM_results.rds	Agree to 1.1e-13
393 comparisons against CLM_scatter_df.rds	Agree to 1.4e-13
27 joint F tests against CLM_f_tests_df.rds	Agree to 6.4e-14
54 sample sizes against CLM_study_ns_df.rds	Identical
Table 1 and Table 2 against the published pages	296 of 296 cells
Appendix estimation cells against the published appendix	3,935 of 3,935 cells
run_all.R run twice	Byte-identical across every output file
Coverage gate over in_text_claims.R	46 of 46 claims printed and matching

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13 R Environment

Table 12: Package versions under R version 4.6.0 (2026-04-24)

Package	Version
tidyverse	2.0.0
dplyr	1.2.1
tidyr	1.3.2
readr	2.2.0
purrr	1.2.2
stringr	1.6.0
ggplot2	4.0.3
estimatr	1.0.6
broom	1.0.13
deming	1.4.1
lmtest	0.9.40
here	1.0.2
knitr	1.51
kableExtra	1.4.0